

Solución tarea 10

PRACTICA BEELINE

1. Configuración de Seguridad en el Nodo Maestro (ultima property)

Se pone false para que se pueda suplantar la identidad sin requerir autenticación.

```
GNU nano 8.6 /home/kali/Desktop/hive4/conf/hive-site.xml
#xml version="1.0" encoding="UTF-8" standalone="no"?
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>

  <property>
    <name>system:java.io.tmpdir</name>
    <value>/tmp/hive/java</value>
  </property>

  <property>
    <name>system:user.name</name>
    <value>kali</value>
  </property>

  <property>
    <name>hive.metastore.warehouse.dir</name>
    <value>/user/hive/warehouse</value>
  </property>

  <property>
    <name>javax.jdo.option.ConnectionURL</name>
    <value>jdbc:derby:/home/kali/Desktop/hive4/metastore_db;create=true</value>
  </property>

  <property>
    <name>javax.jdo.option.ConnectionDriverName</name>
    <value>org.apache.derby.jdbc.EmbeddedDriver</value>
  </property>

  <property>
    <name>hive.server2.enable.doAs</name>
    <value>false</value>
  </property>

</configuration>
```

2. Configuramos el Nodo3

```

ls -lh $(find /home/kali/Desktop/hive-nod03/home/kali/Desktop/
spring-context-5.3.21.jar      100% 124568 16.9MB/s  0:00
jersey-client-1.19.jar        100% 13180 18.4MB/s  0:00
c4lib-2.2.2.jar               100% 7680 5.17MB/s   0:00
log4j-core-2.18.0.jar         100% 181888 15.0MB/s  0:00
jre-impl-10.10.0.jar          100% 6568 4.2MB/s    0:00
jetty-jodi-9.4.45.v20220203.jar 100% 4658 7.0MB/s   0:00
drowizard-metrics-hadoop-metrics2-reporter-0.1.2.jar 100% 1648 3.9MB/s   0:00
spring-core-5.3.21.jar        100% 145900 26.9MB/s  0:00
lithrift-0.16.0.jar           100% 32768 25.3MB/s  0:00
javax.json-api-1.0.jar         100% 2960 12.3MB/s   0:00
netty-codec-native_1.7.7.Final.jar 100% 1388 19.9MB/s  0:00
hive-standalone-metastore-common-4.0.0.jar 100% 1088 22.0MB/s  0:00
jetty-io-9.5.0.jar            100% 6880 28.3MB/s  0:00
STW4-0.0.4.jar               100% 2318 28.3MB/s  0:00
commons-logging-1.2-log-2.18.0.jar 100% 16280 29.1MB/s  0:00
accumulo-start-1.10.1.jar      100% 2168 9.4MB/s    0:00
common-dbgcp-2.0.0.jar         100% 2068 18.0MB/s  0:00
commons-io-2.12.0.jar          100% 6636 19.7MB/s  0:00
curator-framework-5.2.0.jar    100% 3296 8.2MB/s    0:00
jetty-runner-9.4.45.v20220203.jar 100% 7736 23.5MB/s  0:00
arime-memory-core-12.0.0.jar    100% 1088 9.9MB/s    0:00
hive-hcatalog-server-extensions-6.0.0.jar 100% 8768 11.0MB/s  0:00
opencsv-2.3.jar               100% 3960 5.3MB/s    0:00
flatbuffers-java-1.12.0.jar     100% 6360 7.7MB/s    0:00
jline-reader-3.12.1.jar         100% 1476 13.3MB/s  0:00
gpc-api-1.51.0.jar             100% 2556 9.4MB/s    0:00
hive-serde-4.0.0.jar           100% 9648 18.0MB/s  0:00
orc-core-1.8.5.jar             100% 11880 26.0MB/s  0:00
facebookservice.py            100% 568 19.4MB/s   0:00
_init_.py                      100% 837 289.7KB/s  0:00
types.py                       100% 556 163.1KB/s   0:00
facebookbase.py               100% 1917 583.2KB/s  0:00
constants.py                  100% 158 43.5KB/s   0:00
facebookservice-remote        100% 3882 1.0MB/s    0:00
_init_.py                      100% 818 321.6KB/s  0:00
fd303_sample_mgmt.py          100% 5961 1.6MB/s    0:00
TCons.py                      100% 1266 557.2KB/s  0:00
_init_.py                      100% 837 365.3KB/s  0:00
thriftpy.py                   100% 2176 487.9KB/s  0:00
fastbinary.c                  100% 2088 5.3MB/s   0:00
_init_.py                      100% 863 259.5KB/s  0:00
Textool.py                    100% 3994 762.4KB/s  0:00
BinaryProtocol.py             100% 6475 551.9KB/s  0:00
_init_.py                      100% 818 95.6KB/s    0:00
HttpServer.py                 100% 2301 427.6KB/s  0:00
ThonBockingServer.py          100% 1168 1.4MB/s   0:00
server.py                     100% 8157 789.8KB/s  0:00
_init_.py                      100% 807 176.2KB/s  0:00
types.py                      100% 818 168.9KB/s  0:00
types.py                      100% 1000 2.3MB/s   0:00

```

3. Nos conectamos al nodo 3 e iniciamos beeline

```
\\[e]0;\\u0\\h: \\w\\a\\j\\[033[;32m\\]—(\\[033[1;34m\\]\\u@\\h\\[033[;32m\\])-(\\[033[0;1m\\]\\w\\[033[;32m\\])\\n\\[033[;32m\\]—\\[033[1;34m\\]$\\[033[0m\\] ssh nodo3
Linux nodo1 6.12.38+kali1-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Feb 25 13:44:22 2026 from 192.168.1.170
(kali@nodo1)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:97:21:c9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.173/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 84662sec preferred_lft 84662sec
    inet6 2a0c:5a83:ca00:5300:ce71:8bae:8ded:ad6/64 scope global noprefixroute
        valid_lft forever preferred_lft forever
    inet6 fe80::a967:5ada:6c55:dc6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@nodo1)-[~]
$ /home/kali/Desktop/hive4/bin/beeline
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hive4/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hive4/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/kali/Desktop/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Beeline version 4.0.0 by Apache Hive
beeline> █
```

4. Comprobamos que la conexión ha sido exitosa

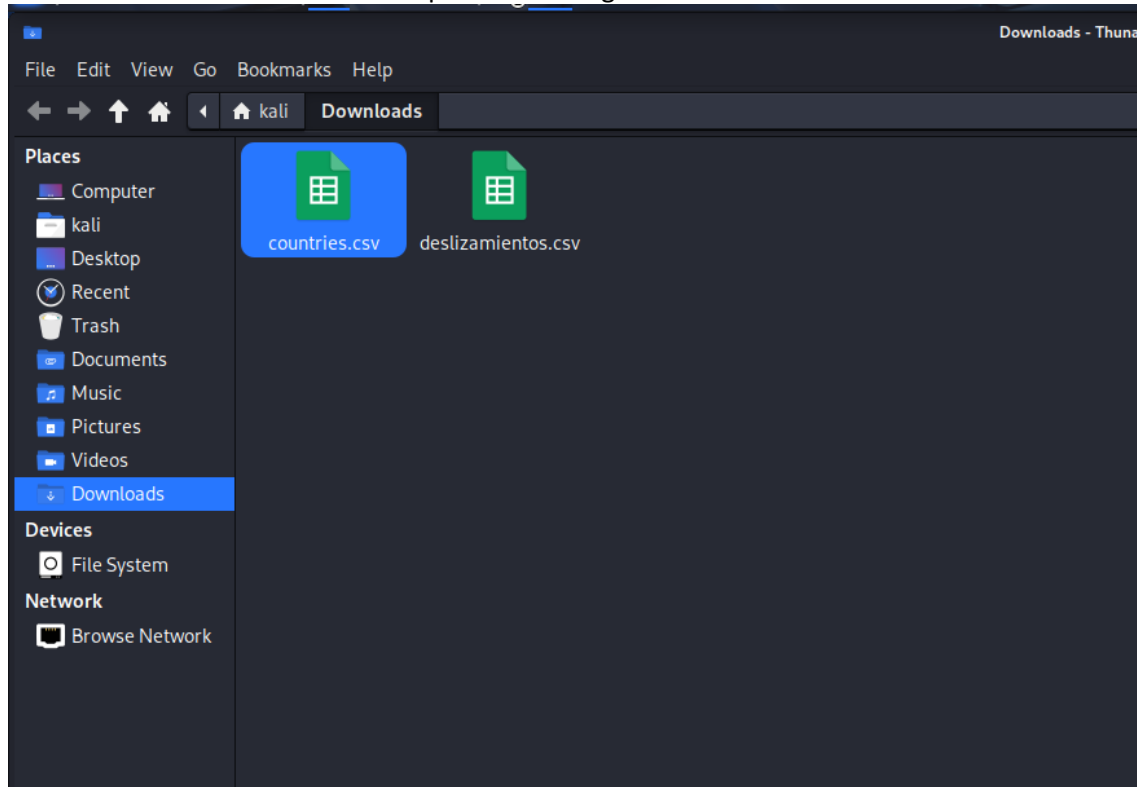
```
0: jdbc:hive2://nodo1:10000> SHOW DATABASES;
INFO : Compiling command(queryId=kali_20260225135904_0df095fc-47a1-4853-8e15-7633c67c64ee): SHOW DATABASES
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:[FieldSchema(name:database_name, type:string, comment:from deserializer)], properties:null)
INFO : Completed compiling command(queryId=kali_20260225135904_0df095fc-47a1-4853-8e15-7633c67c64ee); Time taken: 1.975 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225135904_0df095fc-47a1-4853-8e15-7633c67c64ee): SHOW DATABASES
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260225135904_0df095fc-47a1-4853-8e15-7633c67c64ee); Time taken: 0.12 seconds
+-----+
| database_name |
+-----+
| curso_bigdata |
| default       |
+-----+
2 rows selected (2.787 seconds)
0: jdbc:hive2://nodo1:10000> █
```

PRACTICA NASA

1. Creamos una carpeta dentro de Hadoop llamada nasa

```
(kali@nodo1)~$ /home/kali/Desktop/hadoop/bin/hdfs dfs -mkdir -p /user/hive/nasa
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
```

2. Llevamos los archivos csv a la carpeta de descargas de Linux



3. Iniciamos beeline y creamos la base de datos

```
(kali@nodo1)~$ /home/kali/Desktop/hive4/bin/beeline
Beeline version 4.0.0 by Apache Hive
beeline> !connect jdbc:hive2://nodo1:10000
Connecting to jdbc:hive2://nodo1:10000:
Enter username for jdbc:hive2://nodo1:10000:
Enter password for jdbc:hive2://nodo1:10000:
Connected to: Apache Hive (version 4.0.0)
Driver: Hive JDBC (version 4.0.0)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://nodo1:10000> CREATE DATABASE nasa_db;
INFO : Compiling command(queryId=kali_20260225141020_1aef8318-0d72-41a6-8828-c17fa06cf16b): CREATE DATABASE nasa_db
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225141020_1aef8318-0d72-41a6-8828-c17fa06cf16b); Time taken: 0.072 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225141020_1aef8318-0d72-41a6-8828-c17fa06cf16b): CREATE DATABASE nasa_db
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260225141020_1aef8318-0d72-41a6-8828-c17fa06cf16b); Time taken: 0.421 seconds
No rows affected (0.695 seconds)
0: jdbc:hive2://nodo1:10000>
```

4. Creamos el molde de la tabla

```
No rows affected (0.156 seconds)
0: jdbc:hive2://nodol:10000> CREATE TABLE deslizamientos (
. . . . .> id STRING,
. . . . .> fecha STRING,
. . . . .> hora STRING,
. . . . .> pais STRING,
. . . . .> tipo STRING,
. . . . .> victimas INT
. . . . .> )
. . . . .> ROW FORMAT DELIMITED
. . . . .> FIELDS TERMINATED BY ','
. . . . .> TBLPROPERTIES ("skip.header.line.count"="1");
INFO : Compiling command(queryId=kali_20260225141109_2c6ea2d7-9a2c-402f-8adb-c52741589c5d): CREATE TABLE deslizamientos (
id STRING,
fecha STRING,
hora STRING,
pais STRING,
tipo STRING,
victimas INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
TBLPROPERTIES ("skip.header.line.count"="1")
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225141109_2c6ea2d7-9a2c-402f-8adb-c52741589c5d); Time taken: 0.103 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225141109_2c6ea2d7-9a2c-402f-8adb-c52741589c5d): CREATE TABLE deslizamientos (
id STRING,
fecha STRING,
hora STRING,
pais STRING,
tipo STRING,
victimas INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
TBLPROPERTIES ("skip.header.line.count"="1")
INFO : Starting task [Stage-0:DML] in serial mode
INFO : Completed executing command(queryId=kali_20260225141109_2c6ea2d7-9a2c-402f-8adb-c52741589c5d); Time taken: 0.877 seconds
No rows affected (1.081 seconds)
0: jdbc:hive2://nodol:10000> █
```

5. Cargamos los datos

```
No rows affected (1.081 seconds)
0: jdbc:hive2://nodol:10000> LOAD DATA LOCAL INPATH '/home/kali/Downloads/deslizamientos.csv' INTO TABLE deslizamientos;
INFO : Compiling command(queryId=kali_20260225141219_709a283d-d4e3-4b7b-aed1-70d9b1ee47dc): LOAD DATA LOCAL INPATH '/home/kali/Downloads/deslizamientos.csv' INTO TABLE deslizamientos
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225141219_709a283d-d4e3-4b7b-aed1-70d9b1ee47dc); Time taken: 1.806 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225141219_709a283d-d4e3-4b7b-aed1-70d9b1ee47dc): LOAD DATA LOCAL INPATH '/home/kali/Downloads/deslizamientos.csv' INTO TABLE deslizamientos
INFO : Starting task [Stage-0:MOVE] in serial mode
INFO : Loading data to table nasa_db.deslizamientos from file:/home/kali/Downloads/deslizamientos.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Executing stats task
INFO : Table nasa_db.deslizamientos stats: [numFiles=1, numRows=0, totalSize=3899937, rawDataSize=0, numFilesErasureCoded=0]
INFO : Completed executing command(queryId=kali_20260225141219_709a283d-d4e3-4b7b-aed1-70d9b1ee47dc); Time taken: 4.203 seconds
No rows affected (6.077 seconds)
0: jdbc:hive2://nodol:10000> █
```

6. Filtros de catástrofes

pais	fecha	victimas
India	06/16/2013	5000
Afgghanistan	05/02/2014	2100
China	08/07/2010	1765
Taiwan	08/10/2009	491
Philippines	12/04/2012	430
Brazil	01/12/2011	424
Uganda	03/01/2010	388
Brazil	01/12/2011	378
	10/01/2015	280
China	09/08/2008	277
	08/02/2015	253
	04/28/2015	250
	01/04/2006	240
India	08/06/2010	234
China	05/17/2008	200
Brazil	04/07/2010	196
India	08/06/2010	182
Nepal	08/02/2014	174
India	07/30/2014	151
Indonesia	10/04/2010	145
Bangladesh	06/11/2007	128
Afgghanistan	03/28/2007	114
Brazil	11/24/2008	109
	12/12/2014	108
Philippines	10/09/2009	104
	04/02/2016	104
	05/18/2016	101

7. Agrupación por tipo

tipo	total
Landslide	6657
Mudslide	1828
Rockfall	485
Complex	232
Debris_Flow	173
Other	66
Riverbank_Collapse	28
landslide	21
	18
Unknown	18
Lahar	7
Snow_Avalanche	7
mudslide	7
Translational_Slide	6
Creep	5
Earthflow	3
Rockslide	1
and Netala, all on the Rishikesh-Yamunotri National Highway"	1

18 rows selected (71.357 seconds)

0: jdbc:hive2://nodo1:10000> █

8. Ranking de países

pais	total_incidentes
	3397
United States	1444
India	890
Philippines	548
China	348
Nepal	324
Indonesia	288
Brazil	205
United Kingdom	149
Malaysia	112

10 rows selected (80.961 seconds)
0: jdbc:hive2://nodo1:10000>

9. Creamos la tabla países

```
0: jdbc:hive2://nodo1:10000> CREATE TABLE paises (
    .> nombre STRING,
    .> codigo STRING
    .> )
    .> ROW FORMAT DELIMITED
    .> FIELDS TERMINATED BY ','
    .> TBLPROPERTIES ("skip.header.line.count"="1");
INFO : Compiling command(queryId=kali_20260225143216_616b2d82-1736-433b-bb56-27a71b42b82a): CREATE TABLE paises (
nombre STRING,
codigo STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
TBLPROPERTIES ("skip.header.line.count"="1")
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225143216_616b2d82-1736-433b-bb56-27a71b42b82a); Time taken: 0.013 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225143216_616b2d82-1736-433b-bb56-27a71b42b82a): CREATE TABLE paises (
nombre STRING,
codigo STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
TBLPROPERTIES ("skip.header.line.count"="1")
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260225143216_616b2d82-1736-433b-bb56-27a71b42b82a); Time taken: 0.141 seconds
```

10. Cargamos los datos

```
0: jdbc:hive2://nodo1:10000> LOAD DATA LOCAL INPATH '/home/kali/Downloads/countries.csv' INTO TABLE paises;
INFO : Compiling command(queryId=kali_20260225143221_8659a71e-b49c-45a9-bb1e-d78048976e94): LOAD DATA LOCAL INPATH '/home/kali/Downloads/countries.csv' INTO TABLE paises
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225143221_8659a71e-b49c-45a9-bb1e-d78048976e94); Time taken: 0.035 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225143221_8659a71e-b49c-45a9-bb1e-d78048976e94): LOAD DATA LOCAL INPATH '/home/kali/Downloads/countries.csv' INTO TABLE paises
INFO : Starting task [Stage-0:MOVE] in serial mode
INFO : Loading data to table nasa_db.paises from file:/home/kali/Downloads/countries.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Executing stats task
INFO : Table nasa_db.paises stats: [numFiles=1, numRows=0, totalSize=4120, rawDataSize=0, numFilesErasureCoded=0]
INFO : Completed executing command(queryId=kali_20260225143221_8659a71e-b49c-45a9-bb1e-d78048976e94); Time taken: 0.374 seconds
No rows affected (0.471 seconds)
0: jdbc:hive2://nodo1:10000> DROP TABLE IF EXISTS motivo_pais;
INFO : Compiling command(queryId=kali_20260225143233_afc2c36a-e24b-4f50-877b-e7fa2bde065): DROP TABLE IF EXISTS motivo_pais
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225143233_afc2c36a-e24b-4f50-877b-e7fa2bde065); Time taken: 0.007 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225143233_afc2c36a-e24b-4f50-877b-e7fa2bde065): DROP TABLE IF EXISTS motivo_pais
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260225143233_afc2c36a-e24b-4f50-877b-e7fa2bde065); Time taken: 0.014 seconds
No rows affected (0.052 seconds)
0: jdbc:hive2://nodo1:10000>
```

11. Creamos la tabla motivo_pais

```
0: jdbc:hive2://nodo1:10000>
0: jdbc:hive2://nodo1:10000> CREATE TABLE motivo_pais (
. . . . .>      codigo STRING,
. . . . .>      nombre STRING,
. . . . .>      motivo STRING,
. . . . .>      cantidad INT
. . . . .> )
. . . . .> ROW FORMAT DELIMITED
. . . . .> FIELDS TERMINATED BY ',';
INFO : Compiling command(queryId=kali_20260225143235_fd921ab4-a02b-4768-ac57-eff8943348a5): CREATE TABLE motivo_pais (
codigo STRING,
nombre STRING,
motivo STRING,
cantidad INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225143235_fd921ab4-a02b-4768-ac57-eff8943348a5); Time taken: 0.005 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225143235_fd921ab4-a02b-4768-ac57-eff8943348a5): CREATE TABLE motivo_pais (
codigo STRING,
nombre STRING,
motivo STRING,
cantidad INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260225143235_fd921ab4-a02b-4768-ac57-eff8943348a5); Time taken: 0.092 seconds
No rows affected (0.124 seconds)
```

12. Cargamos los datos

```
0: jdbc:hive2://nodo1:10000> LOAD DATA LOCAL INPATH '/home/kali/Downloads/motivo_pais.csv' INTO TABLE motivo_pais;
INFO : Compiling command(queryId=kali_20260225143353_fa999b26-8304-4cb1-b52e-32e9bae6e5b6): LOAD DATA LOCAL INPATH '/home/kali/Downloads/motivo_pais.csv' INTO TABLE motivo_pais
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260225143353_fa999b26-8304-4cb1-b52e-32e9bae6e5b6); Time taken: 0.039 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260225143353_fa999b26-8304-4cb1-b52e-32e9bae6e5b6): LOAD DATA LOCAL INPATH '/home/kali/Downloads/motivo_pais.csv' INTO TABLE motivo_pais
INFO : Starting task [Stage-0:MOVE] in serial mode
INFO : Loading data to table nasa_db.motivo_pais from file:/home/kali/Downloads/motivo_pais.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Executing stats task
INFO : Table nasa_db.motivo_pais stats: [numFiles=1, numRows=0, totalSize=8352, rawDataSize=0, numFilesErasureCoded=0]
INFO : Completed executing command(queryId=kali_20260225143353_fa999b26-8304-4cb1-b52e-32e9bae6e5b6); Time taken: 0.306 seconds
No rows affected (0.412 seconds)
```

13. Cruzamos las tablas deslizamientos con país

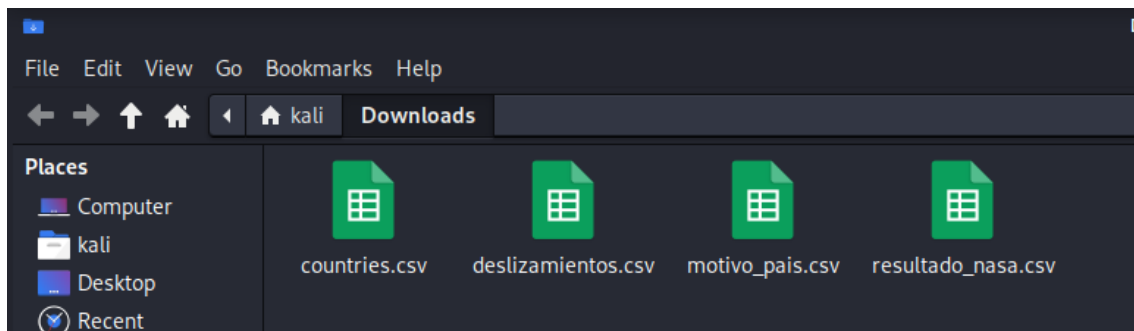
p.codigo	d.pais	d.motivo	d.victimas
IN	India	Downpour	5000
AF	Afghanistan	Continuous_rain	2100
CN	China	Downpour	1765
PH	Philippines	Tropical_Cyclone	430
BR	Brazil	Downpour	424
UG	Uganda	Downpour	388
BR	Brazil	Downpour	378
CN	China	Dam_Embankment_Collapse	277
IN	India	Downpour	234
CN	China	Earthquake	200
BR	Brazil	Downpour	196
IN	India	Downpour	182
NP	Nepal	Continuous_rain	174
IN	India	Continuous_rain	151
ID	Indonesia	Downpour	145
BD	Bangladesh	Monsoon	128
AF	Afghanistan	Flooding	114
BR	Brazil	Continuous_rain	109
PH	Philippines	Tropical_Cyclone	104
CO	Colombia	Downpour	91
PH	Philippines	Tropical_Cyclone	85
CN	China	Snowfall_snowmelt	83
ID	Indonesia	Rain	78
MG	Madagascar	Tropical_Cyclone	73
NP	Nepal	Downpour	72
MX	Mexico	Tropical_Cyclone	71
MM	Myanmar	Rain	70
DO	Dominican Republic	Tropical_Cyclone	68
CN	China	Downpour	66
CN	China	Tropical_Cyclone	66
BR	Brazil	Downpour	63
NP	Nepal	Downpour	60
ID	Indonesia	Rain	60
BR	Brazil	Downpour	56
CN	China	Downpour	53
IN	India	Downpour	52

36 rows selected (88.142 seconds)

```
0: jdbc:hive2://nodo1:10000>
```

14. Exportación de los datos

```
(kali@nodo1)-[~]
$ cat /home/kali/Downloads/resultado_nasa.csv
p.codigo,d.pais,d.motivo,d.victimas
IN,India,Downpour,5000
AF,Afghanistan,Continuous_rain,2100
CN,China,Downpour,1765
PH,Philippines,Tropical_Cyclone,430
BR,Brazil,Downpour,424
UG,Uganda,Downpour,388
BR,Brazil,Downpour,378
CN,China,Dam_Embankment_Collapse,277
IN,India,Downpour,234
CN,China,Earthquake,200
BR,Brazil,Downpour,196
IN,India,Downpour,182
NP,Nepal,Continuous_rain,174
IN,India,Continuous_rain,151
ID,Indonesia,Downpour,145
BD,Bangladesh,Monsoon,128
AF,Afghanistan,Flooding,114
BR,Brazil,Continuous_rain,109
PH,Philippines,Tropical_Cyclone,104
CO,Colombia,Downpour,91
PH,Philippines,Tropical_Cyclone,85
CN,China,Snowfall_snowmelt,83
ID,Indonesia,Rain,78
MG,Madagascar,Tropical_Cyclone,73
NP,Nepal,Downpour,72
MX,Mexico,Tropical_Cyclone,71
MM,Myanmar,Rain,70
DO,Dominican Republic,Tropical_Cyclone,68
CN,China,Downpour,66
CN,China,Tropical_Cyclone,66
BR,Brazil,Downpour,63
NP,Nepal,Downpour,60
ID,Indonesia,Rain,60
BR,Brazil,Downpour,56
CN,China,Downpour,53
IN,India,Downpour,52
```



PRACTICA SUCESOR DE KROOS

1. Creamos un archivo llamado limpiar_datos.py

```
File Edit Search View Document Help
1 import pandas as pd
2
3 # Cargar FBref
4 df_fbref = pd.read_csv('players_data-2024_2025.csv')
5 # Eliminar filas donde el nombre es 'Player' (cabeceras repetidas)
6 df_fbref = df_fbref[df_fbref['Player'] != 'Player']
7 # Corregir edad: de '24-120' a '24'
8 df_fbref['Age'] = df_fbref['Age'].str.split('-').str[0]
9 df_fbref.to_csv('fbref_clean.csv', index=False)
10
11 # Cargar Transfermarkt
12 df_tm = pd.read_csv('players.csv')
13 # Solo nos interesan: name y market_value_in_eur
14 df_tm = df_tm[['name', 'market_value_in_eur']]
15 df_tm.to_csv('tm_clean.csv', index=False)
```

2. Ejecutamos el archivo

```
kali@node0: ~/Desktop
Session Actions Edit View Help

\\[e]o;\\u@h[\\w\\a\\j\\[033[;32m\\]—(\\[\\033[1;34m\\])\\u@\\h\\[\\033[;32m\\])—(\\[\\033[0;1m\\])\\w\\[\\033[;32m\\])\\n\\[\\033[;32m\\]l—(\\[\\033[1;34m\\])$\\[\\033[0m\\] python3 limpiar_datos.py
\\[e]o;\\u@h[\\w\\a\\j\\[033[;32m\\]—(\\[\\033[1;34m\\])\\u@\\h\\[\\033[;32m\\])—(\\[\\033[0;1m\\])\\w\\[\\033[;32m\\])\\n\\[\\033[;32m\\]l—(\\[\\033[1;34m\\])$\\[\\033[0m\\] █
```

3. Creamos la carpeta en HDFS y subimos los archivos

[illegible]

4. Nos conectamos a beeline

```
kali@kali: ~  
Session Actions Edit View Help  
  
[ve]0;|u0|h: \w\|\{033[;32m\} |---(\{033[;34m\}|u0|h\{033[;32m\})- (\{033[0;1m\}|w\{033[;32m\})\n\{033[;32m\} |-(\{033[;34m\}$\{033[0m\} /home/kali/Desktop/hive4/bin/beeline  
beeline> !connect jdbc:hive2://nodol:10000  
Connecting to jdbc:hive2://nodol:10000  
Enter username for jdbc:hive2://nodol:10000:  
Enter password for jdbc:hive2://nodol:10000:  
Connected to: Apache Hive (version 4.0.0)  
Driver: Hive JDBC (version 4.0.0)  
Transaction isolation: TRANSACTION_REPEATABLE_READ  
0: jdbc:hive2://nodol:10000: █
```

```

[*beeline]:u@h: \w[a]\[\033;32m\]—[\033;31;34m\]u@ \h[\033;32m\]—[\033;01m\]w[\033;32m\]\n[\033;32m\]—[\033;31;34m\]h[\033;00m\] /home/kali/Desktop/hive4/bin/beeline
beeline version 4.0.0 by Apache Hive
beeline: connect jdbc:hive2://nodol:10000
Connecting to jdbc:hive2://nodol:10000
Enter username for jdbc:hive2://nodol:10000:
Enter password for jdbc:hive2://nodol:10000:
Connected to Apache Hive (version 4.0.0)
Driver: Hive JDBC (version 4.0.0)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://nodol:10000> DROP TABLE IF EXISTS rendimientio;
INFO : Compiling command(queryId=kali_20260228103115_6110f901-3d06-41ce-b327-115156a52130): DROP TABLE IF EXISTS rendimientio
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldsSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260228103115_6110f901-3d06-41ce-b327-115156a52130): Time taken: 0.031 seconds
INFO : Concurency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260228103115_6110f901-3d06-41ce-b327-115156a52130): DROP TABLE IF EXISTS rendimientio
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260228103115_6110f901-3d06-41ce-b327-115156a52130): Time taken: 0.023 seconds
No rows affected (0.12 seconds)
0: jdbc:hive2://nodol:10000> DROP TABLE IF EXISTS finanzas;
INFO : Compiling command(queryId=kali_20260228103116_c3c20f43-b1a9-42e1-a375-1f366d3b8de9): DROP TABLE IF EXISTS finanzas
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldsSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260228103116_c3c20f43-b1a9-42e1-a375-1f366d3b8de9): Time taken: 0.036 seconds
INFO : Concurency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260228103116_c3c20f43-b1a9-42e1-a375-1f366d3b8de9): DROP TABLE IF EXISTS finanzas
INFO : Starting task [Stage-0:DDL] in serial mode
INFO : Completed executing command(queryId=kali_20260228103116_c3c20f43-b1a9-42e1-a375-1f366d3b8de9): Time taken: 0.01 seconds
No rows affected (0.072 seconds)
0: jdbc:hive2://nodol:10000>
0: jdbc:hive2://nodol:10000> CREATE EXTERNAL TABLE finanzas (
. . . . .-> name STRING,
. . . . .-> market_value DOUBLE
. . . . .-> )
. . . . .-> ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
. . . . .-> STORED AS TEXTFILE LOCATION '/user/scouting/finanzas'
. . . . .-> TBLPROPERTIES ('skip.header.line.count'='1');
INFO : Compiling command(queryId=kali_20260228103116_875c7411-26c4-4305-b288-29fa8e7724b4): CREATE EXTERNAL TABLE finanzas (
name STRING,
market_value DOUBLE
)
ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
STORED AS TEXTFILE LOCATION '/user/scouting/finanzas'
TBLPROPERTIES ('skip.header.line.count'='1')
INFO : Semantic Analysis Completed (retrial = false)
INFO : Created Hive schema: Schema(fieldsSchemas:null, properties:null)
INFO : Completed compiling command(queryId=kali_20260228103116_875c7411-26c4-4305-b288-29fa8e7724b4): Time taken: 0.03 seconds
INFO : Concurency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=kali_20260228103116_875c7411-26c4-4305-b288-29fa8e7724b4): CREATE EXTERNAL TABLE finanzas (

```

```

0: jdbc:hive2://nodo1:10000> SELECT
. . . . .>      r.player AS Jugador,
. . . . .>      r.age AS Edad,
. . . . .>      r.squad AS Equipo,
. . . . .>      f.market_value / 1000000 AS Precio_Millones,
. . . . .>      (r.prpg + r.kp + r.sca) AS Impacto_Total,
. . . . .>      ((r.prpg + r.kp + r.sca) / (f.market_value / 1000000)) AS Ratio_Rentabilidad,
. . . . .>      'Suma(Prog+Clav+Crea)' AS Info_Impacto,
. . . . .>      '← (Impacto / Precio)' AS Info_Ratio
. . . . .> FROM rendimiento r
. . . . .> JOIN finanzas f ON r.player = f.name
. . . . .> WHERE
. . . . .>      f.market_value ≥ 50000000
. . . . .>      AND r.pos LIKE '%MF%'
. . . . .>      AND r.pos NOT LIKE '%FW%'
. . . . .>      AND r.cmp_perc ≥ 75.0
. . . . .>      AND r.age ≤ 24
. . . . .>      AND r.min > 900
. . . . .> ORDER BY Ratio_Rentabilidad DESC
. . . . .> LIMIT 20;
INFO : Compiling command(queryId=kali_20260228103417_a381b8e2-382d-4a12-8a54-cc75b1a39383): SELECT
r.player AS Jugador,
r.age AS Edad,
r.squad AS Equipo,
f.market_value / 1000000 AS Precio_Millones,
(r.prpg + r.kp + r.sca) AS Impacto_Total,
((r.prpg + r.kp + r.sca) / (f.market_value / 1000000)) AS Ratio_Rentabilidad,
'Suma(Prog+Clav+Crea)' AS Info_Impacto,
'← (Impacto / Precio)' AS Info_Ratio
FROM rendimiento r
JOIN finanzas f ON r.player = f.name
WHERE
f.market_value ≥ 50000000
AND r.pos LIKE '%MF%'
AND r.pos NOT LIKE '%FW%'
AND r.cmp_perc ≥ 75.0

```

jugador	edad	equipo	precio_millones	impacto_total	ratio_rentabilidad	info_impacto	info_ratio
Enzo Fernández	23	Chelsea	85.0	436	5.129411764705883	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Morgan Gibbs-White	24	Nott'ham Forest	65.0	330	5.076923076923077	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Xavi Simons	21	RB Leipzig	60.0	299	4.983333333333333	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Manu Kone	23	Roma	50.0	230	4.6	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Elliot Anderson	21	Nott'ham Forest	60.0	267	4.45	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Pedri	23	Barcelona	140.0	595	4.25	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Pablo Barrios	21	Atlético Madrid	60.0	234	3.9	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Dominik Szoboszlai	23	Liverpool	85.0	306	3.6	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Vitinha	24	Paris S-G	90.0	324	3.6	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Carlos Baleba	20	Brighton	60.0	205	3.4166666666666665	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Adam Wharton	20	Crystal Palace	60.0	192	3.2	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Sandro Tonali	24	Newcastle Utd	75.0	234	3.12	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Aleksandar Pavlovic	20	Bayern Munich	55.0	165	3.0	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Andreas Onana	22	Aston Villa	32.0	174	3.4	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Jude Bellingham	21	Real Madrid	160.0	346	2.1625	Suma(Prog+Clav+Crea)	← (Impacto / Precio)
Jamal Musiala	21	Bayern Munich	140.0	259	1.85	Suma(Prog+Clav+Crea)	← (Impacto / Precio)

16 rows selected (91.064 seconds)

0: jdbc:hive2://nodol1:10000